AISCI5000: HW3 (80 points)

Q1 (20 points)

Consider a two-class classification problem in $D = 2$ dimensions where the linear discriminant function is defined as:

$$y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$$

with weight vector $\mathbf{w} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and bias parameter $w_0 = 2$.

a) **(7 points)** Write down the algebraic equation of the decision boundary $y(\mathbf{x}) = 0$. Draw the decision boundary and calculate the unit normal vector $\hat{\mathbf{n}}$ that determines the orientation of this boundary.

b) **(5 points)** Calculate the signed distance r_0 from the decision boundary to the origin $\mathbf{0} = (0, 0)^T$. State on which side of the decision boundary (class $\mathcal{C}_1$ where $y \geq 0$, or class $\mathcal{C}_2$ where $y < 0$) the origin lies.

c) **(8 points)** Given a test observation $\mathbf{x}_A = (2, -1)^T$, determine its assigned class. Calculate the signed distance r_A from $\mathbf{x}_A$ to the decision boundary.

Q2 (20 points)

For a two-class logistic regression model with fixed basis function vector $\boldsymbol{\phi}(\mathbf{x})$, the class posterior is modeled as:

$$y_n = p(\mathcal{C}_1|\boldsymbol{\phi}_n) = \sigma(a_n), \quad \text{where } a_n = \mathbf{w}^T\boldsymbol{\phi}_n \text{ and } \sigma(a) = \frac{1}{1 + e^{-a}}$$

The negative log-likelihood over a dataset of N independent observations $\{\boldsymbol{\phi}_n, t_n\}_{n=1}^N$ with binary targets $t_n \in \{0, 1\}$ is given by the cross-entropy error function:

$$E(\mathbf{w}) = -\sum_{n=1}^{N} \{t_n \ln y_n + (1 - t_n) \ln(1 - y_n)\}$$

a) **(6 points)** Prove that the derivative of the logistic sigmoid function satisfies $\frac{d\sigma(a)}{da} = \sigma(a)(1 - \sigma(a))$.

b) **(14 points)** Using the result from part (a) and the chain rule, prove that the gradient of $E(\mathbf{w})$ with respect to the weight vector $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = \sum_{n=1}^{N} (y_n - t_n)\boldsymbol{\phi}_n$$

c) **(0 points, optional)** Suppose the training data is linearly separable in the feature space $\boldsymbol{\phi}$. Explain why minimizing $E(\mathbf{w})$ via unregularized maximum likelihood leads to severe over-fitting ($\|\mathbf{w}\| \to \infty$).

Q3: (20 points)

In K-class classification, let L_{kj} denote the loss matrix element representing the penalty for assigning a data point whose true class is $\mathcal{C}_k$ to class $\mathcal{C}_j$. The expected loss across all input space $\mathbb{R}^D$ is given by Eq. 10 on slide 19 of lecture 4.

a) **(6 points)** Define a loss matrix L_{kj} where $L_{kk} = 0$ for all k, and $L_{kj} = 1$ for all $k \neq j$. Show that minimizing the expected loss under this loss matrix is mathematically equivalent to minimizing the overall misclassification rate $p(\text{mistake})$.

b) **(14 points)** For a binary classification problem ($K = 2$), express the decision rule that minimizes expected loss in terms of posterior probabilities $p(\mathcal{C}_1|\mathbf{x})$ and $p(\mathcal{C}_2|\mathbf{x})$ under a general loss matrix:

$$\mathbf{L} = \begin{pmatrix} 0 & L_{12} \\ L_{21} & 0 \end{pmatrix}$$

Show that the optimal decision threshold θ for assigning $\mathbf{x}$ to class $\mathcal{C}_1$ (i.e., assign to $\mathcal{C}_1$ if $p(\mathcal{C}_1|\mathbf{x}) > \theta$) is given by:

$$\theta = \frac{L_{21}}{L_{12} + L_{21}}$$

Confirm that when $L_{12} = L_{21}$, this simplifies to the minimum misclassification decision rule $\theta = 0.5$.

[Hint: Do not directly use Eq 10 on slide 19. Instead, consider the argument made for a single input $\mathbf{x}$. What would be the expected loss if a given input $\mathbf{x}$ is assigned to class $\mathcal{C}_1$, and what would be the expected loss if it is assigned to $\mathcal{C}_2$? The optimal decision is made by picking the smaller expected loss between the two.]

Q4 (20 points)

Consider a rare-disease screening dataset of $N = 100,000$ patients, where only $N_{\text{pos}} = 100$ patients actually have the disease (Class $\mathcal{C}_2$, $t = 1$) and $N_{\text{neg}} = 99,900$ patients are healthy (Class $\mathcal{C}_1$, $t = 0$). Let us consider two candidate classifiers evaluated on this dataset:

- **Trivial Classifier A:** Always predicts "Healthy" ($t = 0$) for every patient.
- **Diagnostic Model B:** Correctly identifies 80 true positive patients ($N_{TP} = 80$), misclassifies 20 diseased patients as healthy ($N_{FN} = 20$) and incurs 320 false positives ($N_{FP} = 320$).

a) **(8 points)** Construct the 2×2 Confusion Matrix ($N_{TN}, N_{FP}, N_{FN}, N_{TP}$) for both Model A and Model B. Calculate the classification **Accuracy** for both models. Based solely on Accuracy, which model appears superior?

b) **(12 points)** Calculate the **Precision**, **Recall (Sensitivity)**, and F**-score** for both Model A and Model B.

c) **(0 points, optional)** Explain analytically why Accuracy fails on severely imbalanced datasets when N_{TN} is extremely large, and why the F-score provides an informative and robust assessment in such scenarios.